NZ Government Bond Daily Flows — Method

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This builds daily net NZ$ flows between the NZ Government and private bondholders, 1 Feb 2000 to 30 Sep 2026, from 8 public files in 8 steps. The target is `net_flow_private` in NZ$ millions: positive when the Government pays holders (coupons, maturities, buybacks), negative when holders pay the Government (new bonds). 3,573 of 3,587 bond-months reconcile exactly to NZ Debt Management's published amounts outstanding.

Sources

Download these into one folder called `raw` and never edit them. Write each file's publish date on it, since NZ Debt Management (NZDM) republishes in place.

#	File (sheet used)	Publisher	Feeds
1	Government Bonds – on issue, history at month end (`Month_end`)	NZDM	Bond list, coupons, maturities, month-end holdings by holder (Steps 1–2)
2	Non-resident holdings of individual bonds, D31 (`Data`)	RBNZ	Private and non-resident holdings from Jul 2015 (Step 2)
3	Inflation-indexed bonds – factors spreadsheet (all tabs)	NZDM	Inflation factor Kt per IIB coupon date (Step 3)
4	NZ CPI, quarterly (series NZLCPIALLQINMEI)	OECD via FRED	Kt for the 2016 IIB only, which file 3 omits (Step 3)
5	Government Bonds – Tender Issuance History (`Nominals` , `IIBs`)	NZDM	Bonds sold at tender (Step 6)
6	Government Bonds – Syndication (`Syndication` , `Syndicated Tap`)	NZDM	Bonds sold by syndicate (Step 6)
7	Government Bonds – Repurchases (`Tender` , `LSAP Repurchase`)	NZDM	Buybacks from investors, and from the RBNZ's LSAP holdings (Step 6)

#	File (sheet used)	Publisher	Feeds
8	Non market issuance – issuance history (bond rows only)	NZDM	Bonds issued directly to Crown entities (Step 6)

Out of scope: Treasury bills, Euro-commercial paper and retail Kiwi Bonds.

Steps

Each step produces one sheet in the workbook. Stop at the end of each step and check its stated count before moving on.

1. **Clean the month-end holdings** (file 1 → sheet `holdings`).
   a. Delete every row whose first cell is `-`.
   b. Make a bond key: `GB` for nominal or `IIB` for inflation-indexed, plus maturity as MMYY. For example, GB0427 is the April 2027 nominal bond. RBNZ D31 uses the same key.
   c. Coupon: turn text such as `4.50%` into 0.045. Then give every row of a bond that bond's most common coupon. Jul–Aug 2021 rows carry another bond's coupon; this is where Miles's 3.5% (2032) and 2.0% (2041) came from.
   d. Month: set each date to its calendar month-end, `=EOMONTH(date,0)`. In Aug 2000, Feb 2001 and Nov 2002, all but the first row of the block repeat the previous month's date; relabel the second copy of each bond to the next month. Aug 2009 and Aug 2010 hold extra mid-month snapshots; keep the last one.
   e. Delete GB0532's May 2021 row. That bond first settled on 16 Jun 2021.
   f. Add the two 4.55% Sep 2001 loans (prefixes XL08 and 0008) into one bond.
   g. Check: 321 month-ends (Jan 2000 to Sep 2026) and 43 bonds.
2. **Split holders** (add file 2). Match D31 to each row on month-end and bond.
   - `private` = D31 "Total in the market" where it exists (Jul 2015 on), else NZDM "Market Bonds".
   - `rbnz_secondary` = NZDM "Market Bonds" − `private`. These are RBNZ holdings bought from investors, mostly under the LSAP programme; they peaked at NZ$53.6bn in Apr 2021.
   - `crown` = RBNZ + EQC + SRESL + `rbnz_secondary`.
   - `nonres` = D31 "Non-residents".

3. **Inflation factor Kt** (files 3–4 → sheet `kt_factors`). Stack the tabs of file 3 into one table: bond, coupon date, Kt. File 3 has no tab for IIB0216, the 4.5% Feb 2016 bond, so build it from CPI. Start Kt = 100 on 15 Nov 1995. Each quarter, Kt = previous Kt × (1 + p/100). Here p is the average of the two quarterly CPI % changes ending two quarters before the payment's quarter. Round p and Kt to 2 decimals.
4. **Payment schedule** (sheet `payments`, one row per bond and payment).
   - Scheduled dates: `=EDATE(maturity,-6*n)` for n = 0, 1, 2… back to Feb 2000 (use `-3*n` for IIBs). Each date is a coupon; at n = 0 add a second row for principal.
   - `holdings_month_end` = `=EOMONTH(scheduled_date,-1)`. Keep only rows where the bond has holdings that month.
   - `pay_date` = `=WORKDAY(scheduled_date-1,1,holidays)`. This leaves a business day unchanged and moves a weekend or holiday to the next business day.
   - Face amount: coupon = holdings × coupon ÷ 2 (÷ 4 for IIBs); principal = holdings. Cash: face × Kt/100 for IIBs, using Kt at the scheduled date; face for nominal bonds. Do this separately for `private`, `crown` and `nonres`.
   - Check: 751 rows.
5. **Holiday list** (sheet `holidays`). NZ public holidays plus Wellington Anniversary Day. Business days are weekdays not on this list.
6. **Issuance and buybacks** (files 5–8 → sheet `events`, one row per deal). Amounts are face value; IIB rows are multiplied by the latest Kt on or before settlement, ÷ 100.

Source	Amount column	Settlement date	Sign	Goes to
Tender, `Nominals` and `IIBs`	Volume Accepted	`=WORKDAY(tender_date,3,holidays)`	−	Issuance
Syndication, `Syndication` and `Syndicated Tap` (other tabs repeat these)	Total Amt Issued	Settlement Date	−	Issuance
Repurchase, `Tender`	Volume accepted	`=WORKDAY(tender_date,3,holidays)`	+	Buyback
Repurchase, `LSAP Repurchase`	Volume accepted	Settlement Date	+	Crown memo

Source	Amount column	Settlement date	Sign	Goes to
Non-market issuance, bond rows	Amount, as published (negative = issued)	Settlement Date	as published	Crown memo

7. **Daily table** (sheet `daily_flows`). List every calendar day from 1 Feb 2000 to 30 Sep 2026. Fill each column with `SUMIFS` over `payments` (by `pay_date`) or `events` (by settlement date), filtered as the column map says. `net_flow_private` = the sum of the 8 component columns.
8. **Check** (sheet `checks`). For every bond and month, the change in total outstanding should equal tender + syndication + non-market issuance − buybacks − LSAP repurchases − the amount maturing. Expect 3,573 of 3,587 bond-months to match exactly. Annual coupon and principal totals should match Amar's table to ±NZ$0.05bn.

Column map

All amounts are NZ$ millions. The first 8 columns sum to the target; memo columns sit beside it and are not added in.

Column in `daily_flows`	What it is	Source files	Built in
`nom_coupon`	Coupons on nominal bonds paid to private holders	1, 2	Step 4, summed by `pay_date`
`nom_principal`	Nominal bonds repaid at maturity to private holders	1, 2	Step 4
`nom_issuance`	New nominal bonds sold (negative)	5, 6	Step 6
`nom_buyback`	Nominal bonds bought back from investors	7 (`Tender`)	Step 6
`iib_coupon`	IIB coupons to private holders, face × Kt/100	1, 2, 3, 4	Steps 3–4

Column in `daily_flows`	What it is	Source files	Built in
`iib_principal`	IIB maturities to private holders, face × Kt/100	1, 2, 3, 4	Steps 3–4
`iib_issuance`	New IIBs sold (negative), face × Kt/100	3, 5, 6	Step 6
`iib_buyback`	IIBs bought back from investors, face × Kt/100	3, 7	Step 6
`net_flow_private`	**Target: sum of the 8 columns above**	—	Step 7
`iib_coupon_face`, `iib_principal_face`, `iib_issuance_face`, `iib_buyback_face`	Memo: the IIB columns before inflation adjustment	as above	Steps 4, 6
`nonres_payments`	Memo: coupons + principal to non-residents; blank before Aug 2015	1, 2	Step 4
`crown_payments`	Memo: coupons + principal to RBNZ, EQC, SRESL and RBNZ secondary holdings	1, 2	Step 4
`crown_nonmarket_issuance`	Memo: bonds issued directly to, or redeemed from, Crown entities	8	Step 6
`crown_lsap_repurchase`	Memo: Treasury buying LSAP bonds back from the RBNZ	7 (`LSAP Repurchase`)	Step 6
`nz_business_day`	TRUE if banks are open in Wellington	holidays	Step 5

The workbook's `column_map` sheet repeats this table and adds the columns of `payments` and `events`.

Assumptions

1. **Who gets paid** is read from the month-end before the payment. NZ bonds close their register about 10 days before payment, which falls in that month.
2. **Dates** are the day the cash moves: weekend and holiday payments go to the next Wellington business day. Downstream FX tests should use the next trading day after this date.
3. **Tenders and buyback tenders settle three business days after the tender (T+3).** NZDM's files give no settlement date. T+3 reconciles 3,573 of 3,587 bond-months, against 3,448 for T+2 and 3,459 for T+4.
4. **Issues and buybacks are at face value** (IIBs at face × Kt), not at the price paid. For low-coupon bonds sold at high yields the cash can be about 15% below face.
5. **All tender and syndication volume goes to private holders.** Direct sales to Crown entities are in file 8.
6. **RBNZ secondary holdings before Jul 2015 are zero,** because D31 starts then. D31 shows them near zero until Jun 2017 (NZ$150m in one month).
7. **The RBNZ's LSAP purchases from investors are left out of the daily columns.** They are monetary operations, not Government debt flows, and are not published by day.
8. **First coupons are paid in full.** Accrued interest paid by buyers at issue is ignored.
9. **The 2016 IIB's Kt is an estimate:** the CPI formula reproduces NZDM's published Kt to ±0.01, but its base date of 15 Nov 1995 is assumed.

Checks and known gaps

The workbook's `checks` sheet holds every result below.

Check	Result
Month-end outstanding vs dated flows	3,573 of 3,587 bond-months exact; 14 breaks totalling NZ$2.4bn
Annual coupons and principal vs Amar's table	Within NZ$0.05bn every year, 2000–2026

Check	Result
Kt rebuilt from CPI vs NZDM's published Kt	Within 0.01 on all 34 testable dates
Flows dated on a weekend or holiday	0

Gap	Effect
No buyback records before 2009	About NZ$1.8bn of 2000–2002 pre-maturity reductions in GB0201 and GB0302 cannot be dated; they are the main reconciliation breaks
Non-resident split starts Jul 2015	`nonres_payments` is blank before Aug 2015
GB0433 alternates +NZ$150m / −NZ$150m, Jul–Oct 2017	Noise in NZDM's file; it moves the Oct 2017 coupon by about NZ$2.6m
2016 IIB Kt is estimated	Adds NZ$606m to the 15 Feb 2016 maturity (NZ$1,777m vs NZ$1,171m face)

Credits and sources

Amar set the schedule method: coupon dates stepped back from maturity, holdings taken from the month-end before payment, and the non-resident split from RBNZ D31. His annual totals are the benchmark in Step 8. **Miles** set the exclusion of Crown-held bonds and the event-study framing of payment days, which motivate the date convention in Assumption 2.

Added here: the RBNZ's LSAP holdings removed from private flows, issuance and buybacks for a net daily flow, inflation-adjusted IIB cash, business-day dating, the T+3 settlement test, and the month-by-month reconciliation.

- NZDM investor data: files 1, 3, 5–8, published 14 Jul–7 Oct 2026
- RBNZ D31, non-resident holdings of individual bonds: published 16 Sep 2026
- OECD NZ CPI via FRED, NZLCPIALLQINMEI: last updated 15 May 2025